

SOFTWARE REQUIREMENT SPECIFICATION

(SRS)

APPLICATION NAME: VIDYA SAATHI

AI-Powered Personalized Learning Companion for Students

Application Name	Domain
Vidya Saathi	EdTech Mobile & Web Generative AI

Team Details

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1. Introduction

1.1 Purpose

Vidya Saathi is a 24/7 AI-powered personalized learning companion designed for Indian students (Class 9–12 and Undergraduate) who lack access to affordable, individualized academic support. This document specifies the user requirements, functional modules, and system behavior for the Phase 1 (MVP) build of Vidya Saathi, to be used as the baseline for design and development.

1.2 Scope

This SRS covers Phase 1 functionality only, as committed in the finalized Idea Deck: AI Doubt Resolution, Subject & Topic Selection, Quiz Generation & Assessment, Adaptive Learning Path, and Hindi + Telugu language support. Phase 2 (additional languages, subjects, analytics) and Phase 3 (teacher dashboard, institutional features, offline mode) are explicitly out of scope for this version and are listed separately under Section 4 (Future Phases) so that Phase 1 requirements remain frozen and unambiguous.

1.3 Target Users (Actors)

- Student (Class 9–12 / UG) – the primary direct user; asks doubts, takes quizzes, receives personalized topic recommendations based on their performance
- System Administrator – manages subject/topic master data and monitors platform health.

1.4 Assumptions & Constraints

- Students have access to a smartphone/laptop with internet connectivity (low-bandwidth tolerant).
- AI responses are generated via external model API; response quality depends on prompt design and model availability.
- Phase 1 supports English, Hindi, and Telugu only; content is restricted to Maths, Science (includes Physics, Chemistry, etc), English, and Social Studies.
- No payment/subscription module is included in Phase 1 — the MVP is offered as a free tier.
- User data (profile, doubts, quiz scores) must be stored securely and used only to personalize the individual student's learning path.

2. Modules

1. Registration & Profile Management

Enables a student to self-register, verify identity, log in securely, and manage their profile — the entry point to the platform.

Functionality

- Self-registration by student using name, class, board, phone/email, and password.
- OTP or email-link based account verification/activation.
- Login authentication using username/email and password.
- Forgot password and password reset flow.
- View profile; request to edit profile details.
- Set/change preferred language (Hindi / Telugu / English) at registration and anytime after.
- Account deactivation/reactivation (self-service or by admin in case of misuse).

Provisioning

User

- User ID
- First Name
- Last Name
- Gender
- Date of Birth
- Class/Grade
- Board (CBSE/State/ICSE)
- Username
- Password (hashed)
- Email
- Phone Number
- Preferred Language
- Status (Active/Inactive)
- Security Question & Answer

Class

- Class ID
- Class Name (9, 10, 11, 12, UG-Year1...)

Board

- Board ID
- Board Name

Language

- Language ID
- Language Name
- Language Code

Queries

- How many students are registered (total / by class / by board / by language)?
- How many users are currently active vs. inactive?
- How many users have requested a profile edit?
- What languages are available for selection?

Alerts

- Mandatory-field alert if required registration fields are left blank.
- Registration success / failure alert.
- OTP/verification-link sent and verification-success alerts.
- Invalid username or password alert on login failure.

Reports

- New registrations trend (daily/weekly/monthly).
- Registration distribution by class, board, and preferred language.

2. Subject & Topic Selection

Lets the student narrow the AI's focus to a specific subject and topic/chapter so that doubt answers, quizzes, and recommendations stay relevant.

Functionality

- Display Phase 1 subjects: Maths, Science, English, Social Studies.
- Allow student to select a subject and drill down into topics/chapters.
- Allow switching subject/topic mid-session without losing context.
- Remember the last-used subject/topic for quick resume.

Provisioning**Subject**

- Subject ID

- Subject Name

Topic

- Topic ID
- Subject ID
- Topic Name
- Class/Grade applicability

Queries

- What subjects are available?
- What topics exist under a selected subject?
- Which subject/topic is most frequently selected?

Alerts

- Prompt to select a subject/topic before submitting a doubt, if none is chosen.

Reports

- Subject-wise usage report (selections per day/week).
- Topic-wise demand report highlighting most-requested chapters.

3. Doubt Resolution (AI Companion)

The core AI module — the heart of Vidya Saathi's 24/7 "doubt to clarity" promise.

Functionality

- Student submits a doubt/question (text) in their preferred language.
- AI (LLM) interprets the query and generates a clear, example-based explanation — like a patient tutor available anytime.
- Student can ask follow-up/clarifying questions within the same conversation thread.
- Student can bookmark/save a doubt for later revision.
- Student rates each answer as Helpful / Not Helpful to improve future responses.
- View past doubt-resolution history, searchable by subject/topic/date.

Provisioning**Doubt**

- Doubt ID
- Student ID
- Subject ID

- Topic ID
- Question Text
- AI Answer Text
- Language
- Timestamp
- Helpful Rating

Queries

- How many doubts were resolved today / this week / total?
- What are the most frequently asked doubts per subject/topic?
- What is the average helpfulness rating of AI answers?

Alerts

- Escalation alert when the AI cannot answer with sufficient confidence (suggest rephrasing or contacting a teacher).
- Content-moderation alert for out-of-syllabus or inappropriate queries.

Reports

- Doubt volume report (daily/weekly trend, by subject).
- Subject-wise doubt distribution report.
- AI answer-quality trend report based on student ratings over time.

4. Quiz Generation & Assessment

Converts passive doubt-resolution into active recall — the bridge between answering a doubt and proving mastery, and the data source that powers the Adaptive Learning Path.

Functionality

- AI auto-generates MCQs for the student's selected topic and difficulty level.
- Student attempts the quiz; system captures each answer.
- Instant feedback is shown per question with a short explanation — not just right/wrong.
- System calculates and displays the overall quiz score on completion.
- Quiz difficulty adapts based on the student's prior performance on that topic.

Provisioning

Quiz

- Quiz ID
- Topic ID

- Difficulty Level

Question

- Question ID
- Quiz ID
- Question Text
- Option A–D
- Correct Option
- Explanation Text

Quiz Attempt

- Attempt ID
- Student ID
- Quiz ID
- Score
- Time Taken
- Date

Queries

- How many quizzes has a student attempted, and with what scores?
- What is the average score by topic/subject across all students?
- Which topics show the highest rate of incorrect answers (weak topics)?

Alerts

- Quiz-completion alert with score summary.
- Low-score alert suggesting revision of the topic.
- Milestone/streak alert (e.g., “5 quizzes completed this week”).

Reports

- Score-trend report per student over time.
- Topic-wise accuracy report (correct vs. incorrect attempts).
- Comparative performance report across subjects for a given student.

5. Adaptive Learning Path Management

Closes the Doubt → Quiz → Adapt feedback loop that differentiates Vidya Saathi from static content platforms — it personalizes what each student should study next.

Functionality

- Analyze quiz results and doubt-frequency patterns to identify weak topics per student.
- Recommend the next topic, doubt-revision item, or quiz based on weak-area analysis.
- Continuously update the student's learning path as new quiz/doubt data arrives.
- Display a mastery level (e.g., Beginner / Improving / Mastered) per topic.

Provisioning**Learner Profile**

- Student ID
- Topic ID
- Mastery Score
- Attempts Count
- Last Updated Timestamp

Recommendation Rule

- Rule ID
- Trigger Condition (e.g., score < 50%)
- Recommended Action

Queries

- What is a student's current recommended learning path?
- What are a student's weak topics right now?
- What is the mastery percentage by subject for a given student?

Alerts

- Mastery-achieved alert when a topic crosses the mastery threshold.
- Stagnant-progress alert if a weak topic shows no improvement over multiple attempts.

Reports

- Personalized progress report per student.
- Subject-wise mastery heatmap.
- Improvement-trend report over the study period.

6. Language Localization Management

Ensures Vidya Saathi communicates in the student's most comfortable language — a core differentiator versus English-only platforms, and the basis for the deck's Hindi + Telugu commitment.

Functionality

- Allow student to set/change preferred language for the UI and AI responses.
- Generate doubt-resolution answers, quiz content, and feedback in the chosen language.
- Fall back gracefully to English if a requested language/content combination is unavailable.

Provisioning**Language**

- Language ID
- Language Name
- Language Code

User Language Preference

- Student ID
- Language ID

Queries

- How many users prefer each supported language?
- Which subjects/topics have the highest non-English usage?

Alerts

- Fallback notice if AI content is not yet available in the selected language for a specific topic.

Reports

- Language-usage distribution report across the active user base.

3. Non-Functional Requirements

Category	Requirement
Performance	AI doubt-resolution response within 3–5 seconds under normal load.
Availability	24/7 uptime target of 99% for the doubt-resolution and quiz services.
Scalability	Architecture must support growth from pilot (hundreds) to 90M+ addressable learners without redesign.
Usability	Zero learning curve — chat-app-like interface familiar to students.
Security	Encrypted password storage, secure session handling, role-based access for admin functions.
Data Privacy	Student data used solely for personalizing that student's own learning path; no third-party sharing.
Compatibility	Must work on low-end Android smartphones and low-bandwidth mobile data connections.
Localization	Full UI and AI-response support for English, Hindi, and Telugu in Phase 1.

4. Out of Scope — Future Phases

4.1 Phase 2 (Months 3–6)

- Additional language support: Tamil, Kannada, Bengali.
- Additional subject categories: Commerce, Engineering basics.
- Detailed topic-mastery progress analytics.

4.2 Phase 3 (Months 7–12)

- Teacher Dashboard – class-level insights and weak-topic alerts.
- Monitoring & analytics for training organizations.
- Institutional partnerships — colleges and coaching centres.
- Offline mode for low-bandwidth environments.